

Basic Networking Skills for Digital Forensics

Idriss Muhammad Abdullahi

2025/FWSD/11475

Fellowship in Web Application Security & Digital Forensics

Aminu Idris

17/09/2026

Lab Scenario

You are validating whether a Linux host-based firewall correctly blocks HTTP traffic from one authorized lab client while allowing another. You will capture baseline traffic, apply a narrowly scoped iptables rule, capture the blocked connection behaviour, prove the rule hit using counters, remove the rule and document the evidential differences.

1. Objective

The objective of this lab was to validate that a Linux host-based firewall could selectively block HTTP traffic from a specific client while continuing to allow access from authorized hosts. Network captures and firewall counters were used as forensic evidence.

2. Baseline Testing

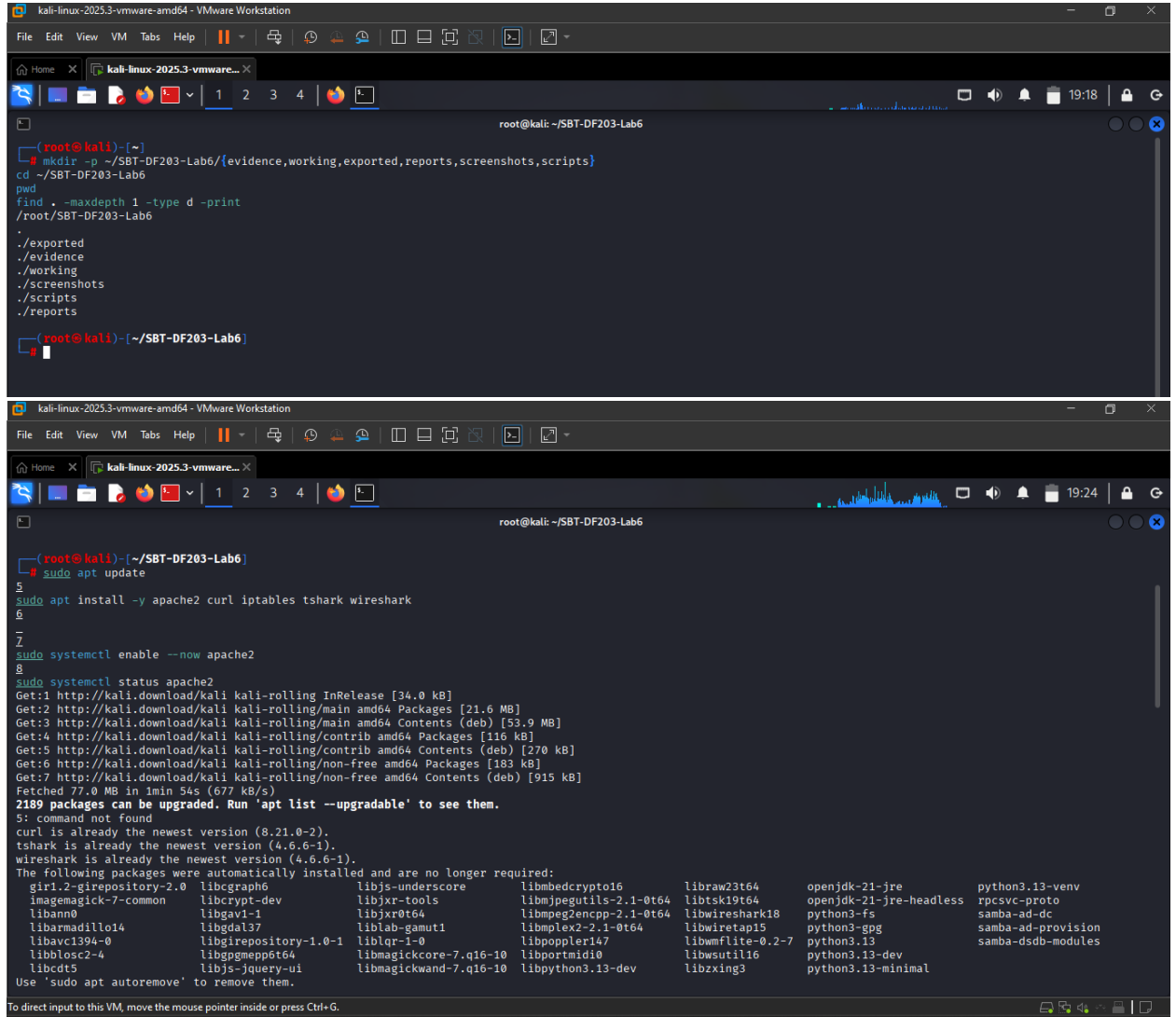
Apache2 was configured and verified to listen on TCP port 80. An authorized client successfully accessed the webpage `firewall_lab.html` using `curl`. Packet capture confirmed a normal TCP three-way handshake followed by HTTP GET and HTTP 200 OK responses.

3. Firewall Rule iimplementation

A DROP rule was inserted at the top of the INPUT chain targeting the assigned blocked client IP address

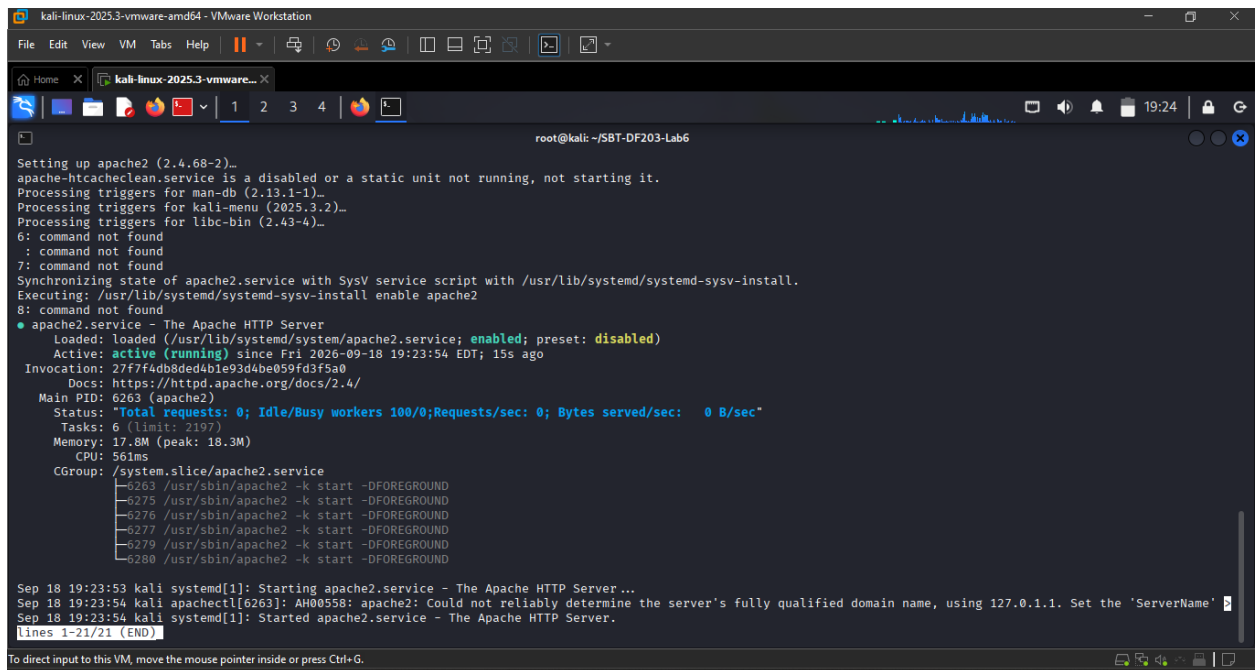
4. Block Traffic Analysis

1. Lab Folder Creation Structure and Evidence Preparation



```
(root@kali)-[~]
└─$ mkdir -p ~/SBT-DF203-Lab6/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab6
pwd
/root/SBT-DF203-Lab6
find . -maxdepth 1 -type d -print
./exported
./evidence
./working
./screenshots
./scripts
./reports
(root@kali)-[~/SBT-DF203-Lab6]
└─$
```

```
(root@kali)-[~/SBT-DF203-Lab6]
└─$ sudo apt update
5
└─$ sudo apt install -y apache2 curl iptables tshark wireshark
6
└─$
7
└─$ sudo systemctl enable --now apache2
8
└─$ sudo systemctl status apache2
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.9 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 77.0 MB in 1min 54s (677 kB/s)
2189 packages can be upgraded. Run 'apt list --upgradable' to see them.
5: command not found
curl is already the newest version (8.21.0-2).
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  gir1.2-girepository-2.0  libcgraph6  libjs-underscore  libmbcrypto16  libraw23t64  openjdk-21-jre  python3.13-venv
  imagemagick-7-common  libcrypt-dev  libjxr-tools  libmjpegutils-2.1-0t64  libtsk19t64  openjdk-21-jre-headless  rpcsvc-proto
  libann0  libgavi-1  libjxr0t64  libmpeg2encpp-2.1-0t64  libwireshark18  python3-fs  samba-ad-dc
  libarmadillo14  libgdal37  liblab-gamut1  libmpeg2-2.1-0t64  libwiretap15  python3-gpg  samba-ad-provision
  libavc1394-0  libgirepository-1.0-1  liblqr-1-0  libpoppler147  libmflite-0.2-7  python3.13  samba-dsdb-modules
  libblosc2-4  libgpgmepp6t64  libmagiccore-7.q16-10  libportmidi0  libwsutil16  python3.13-dev
  libcdt5  libjs-jquery-ui  libmagicwand-7.q16-10  libpython3.13-dev  libzxing3  python3.13-minimal
Use 'sudo apt autoremove' to remove them.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

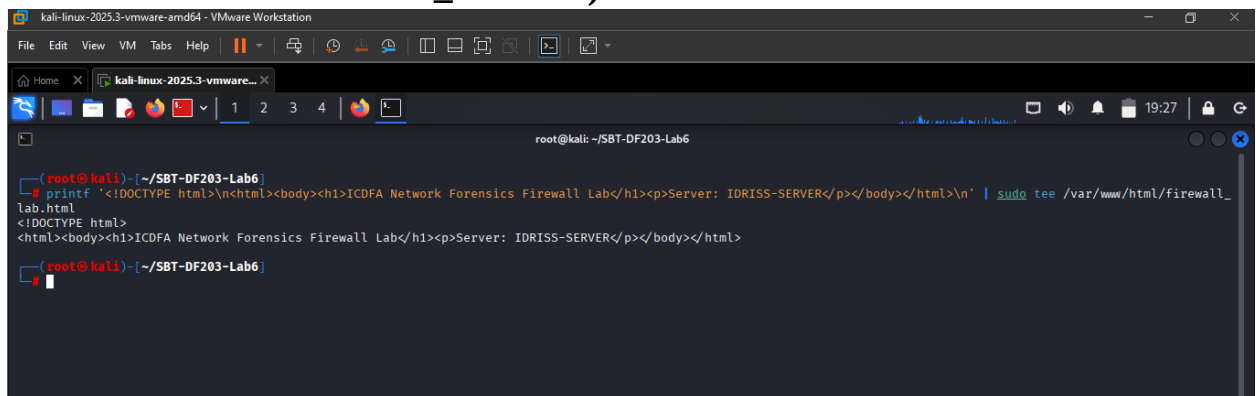


```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

Setting up apache2 (2.4.68-2)...
apache-htcacheclean.service is a disabled or a static unit not running, not starting it.
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2025.3.2)...
Processing triggers for libc-bin (2.43-4)...
6: command not found
: command not found
7: command not found
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
8: command not found
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-09-18 19:23:54 EDT; 15s ago
 Invocation: 27f7f4db8ded4b1e93d4be059fd3f5a0
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 6263 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:  0 B/sec"
     Tasks: 6 (limit: 2197)
  Memory: 17.8M (peak: 18.3M)
     CPU: 561ms
    CGroup: /system.slice/apache2.service
            └─6263 /usr/sbin/apache2 -k start -DFOREGROUND
              └─6275 /usr/sbin/apache2 -k start -DFOREGROUND
                └─6276 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─6277 /usr/sbin/apache2 -k start -DFOREGROUND
                    └─6279 /usr/sbin/apache2 -k start -DFOREGROUND
                      └─6280 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 18 19:23:53 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
Sep 18 19:23:54 kali apachectl[6263]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName'
Sep 18 19:23:54 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-21/21 (END)
```

2. Created test webpage using (printf '<!DOCTYPE html>\n<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: IDRISS-SERVER</p></body></html>\n' | sudo tee /var/www/html/firewall_lab.html)



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

(root@kali)~[~/SBT-DF203-Lab6]
# printf '<!DOCTYPE html>\n<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: IDRISS-SERVER</p></body></html>\n' | sudo tee /var/www/html/firewall_lab.html
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: IDRISS-SERVER</p></body></html>
(root@kali)~[~/SBT-DF203-Lab6]
```

3. Original Firewall State and Hashing results
Sha256sum =
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b85
5

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

(root@kali)~/SBT-DF203-Lab6
# sudo iptables-save | tee reports/iptables_before.rules

(root@kali)~/SBT-DF203-Lab6
# sudo iptables -L -n -v --line-numbers | tee reports/iptables_before.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination

(root@kali)~/SBT-DF203-Lab6
# sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 reports/iptables_before.rules

(root@kali)~/SBT-DF203-Lab6
#
```

Part A: Recording the Lab Network and Baseline Access

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

(root@kali)~/SBT-DF203-Lab6
# sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 reports/iptables_before.rules

(root@kali)~/SBT-DF203-Lab6
# ip -br address | tee reports/server_interfaces.txt
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.126.131/24 fe80::5a87:c64f:ac28:600a/64

(root@kali)~/SBT-DF203-Lab6
# ip route | tee reports/server_routes.txt
default via 192.168.126.2 dev eth0 proto dhcp src 192.168.126.131 metric 100
192.168.126.0/24 dev eth0 proto kernel scope link src 192.168.126.131 metric 100

(root@kali)~/SBT-DF203-Lab6
# sudo ss -ltnp | grep ':80' | tee reports/apache_listener.txt~
LISTEN 0 511 *:80 *:~ users:(("apache2",pid=1019,fd=4),("apache2",pid=1018,fd=4),("apache2",pid=1017,fd=4),("apache2",pid=1016,fd=4),("apache2",pid=1015,fd=4),("apache2",pid=900,fd=4))

(root@kali)~/SBT-DF203-Lab6
#
```

the above network configuration confirms that the Apache web server was running and accessible on port 80.

Test baseline access from allowed client:

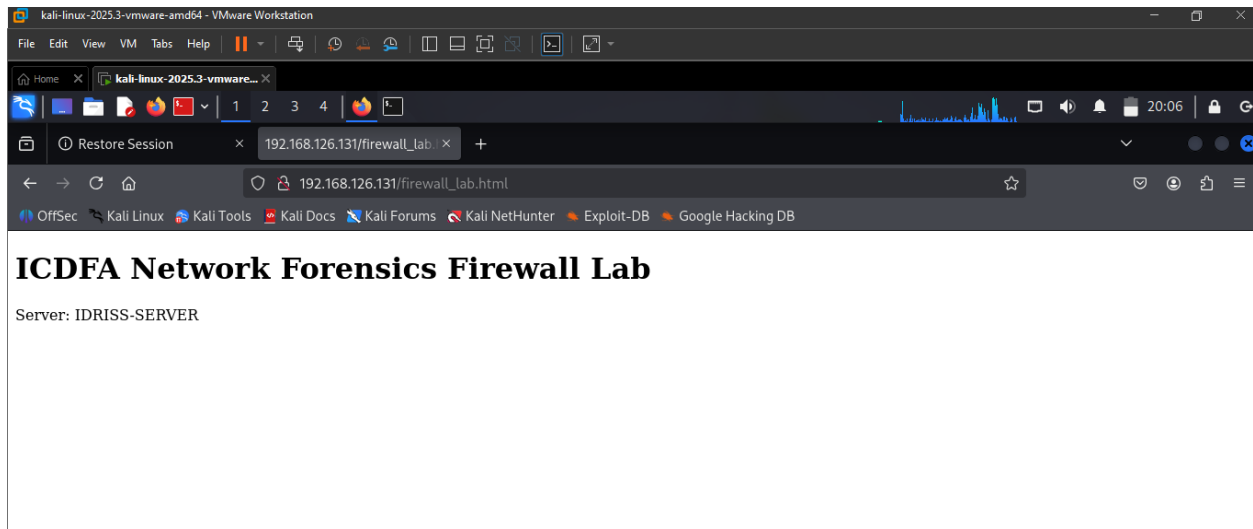
A screenshot of a Kali Linux virtual machine window titled "kali-linux-20253-virtual-machine-amd64 - VM Workstation". The terminal shows a root prompt at "/SBT-DP203-Lab6". A user has entered a curl command to fetch a file from 192.168.126.131. The output shows a successful HTTP GET request with status 200 OK, headers including Date, Server, Last-Modified, ETag, Accept-Ranges, Content-Length, Vary, and Content-Type, and a body containing HTML code for a network forensics lab exercise.

```
kali@kali:~$ curl -v --connect-timeout 5 http://192.168.126.131/firewall_lab.html 2>&1 | tee baseline_curl.txt
* Trying 192.168.126.131:80...
* Established connection to 192.168.126.131 (192.168.126.131 port 80) from 192.168.126.131 port 35498
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0      0  0* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.126.131
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 18 Sep 2026 23:52:34 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 23:31:59 GMT
< ETag: "74-65bca4e8499ee"
< Accept-Ranges: bytes
< Content-Length: 116
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [116 bytes data]
100  116 100   116    0     0 17810    0      0
* Connection #0 to host 192.168.126.131:80 left intact
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: IDRISS-SERVER</p></body></html>
```

```
(root@kali)-[~/SBT-DF203-Lab6]
# curl http://192.168.126.131/firewall_lab.html

<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: IDRIS-SERVER
/p></body></html>

(root@kali)-[~/SBT-DF203-Lab6]
#
```



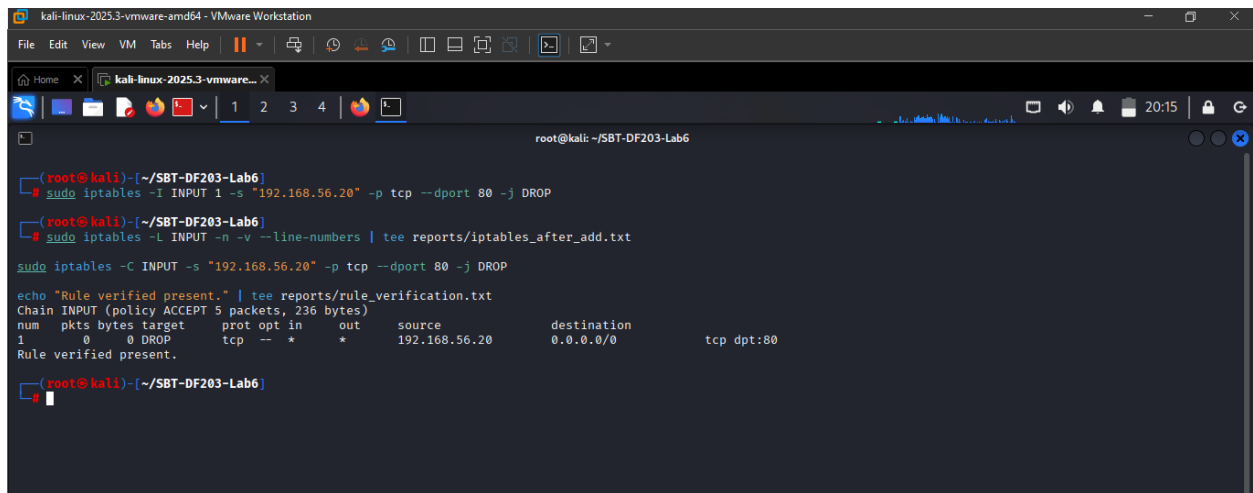
After Capturing, Sha256sum is;

12cbf7252e0257f590691aba5460483fedab4da7438249843aa92caca0c54427

```
(root@kali)-[~/SBT-DF203-Lab6]
# sha256sum evidence/http_allowed.pcapng | tee reports/http_allowed_sha256.txt
12cbf7252e0257f590691aba5460483fedab4da7438249843aa92caca0c54427  evidence/http_allowed.pcapng

(root@kali)-[~/SBT-DF203-Lab6]
#
```

Part C - Applying and Verifying the Blocking Rule



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

(root@kali)~/SBT-DF203-Lab6
# sudo iptables -I INPUT 1 -s "192.168.56.20" -p tcp --dport 80 -j DROP

(root@kali)~/SBT-DF203-Lab6
# sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_add.txt

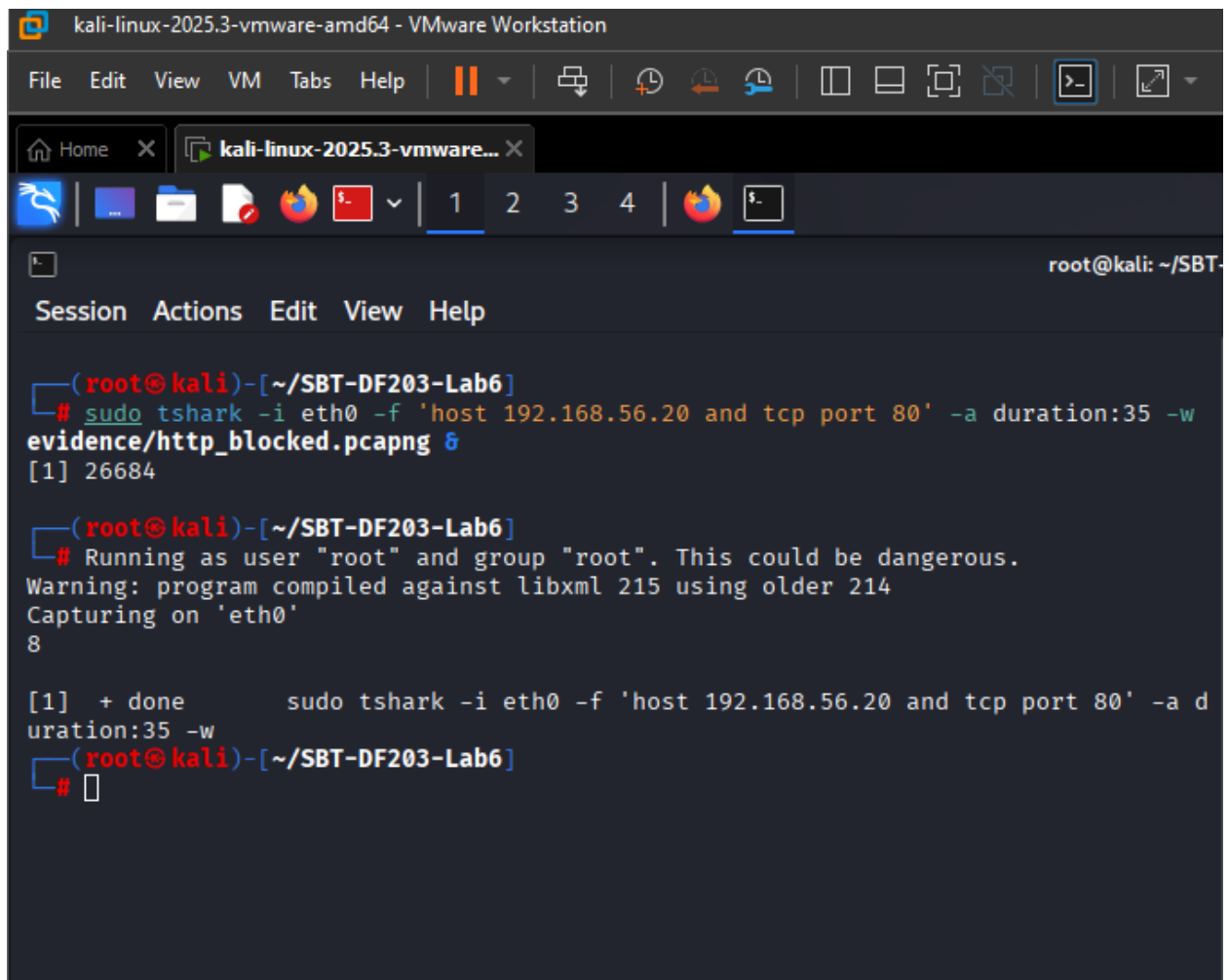
sudo iptables -C INPUT -s "192.168.56.20" -p tcp --dport 80 -j DROP

echo "Rule verified present." | tee reports/rule_verification.txt
Chain INPUT (policy ACCEPT 5 packets, 236 bytes)
num  pkts bytes target    prot opt in     out     source            destination
1      0      0 DROP      tcp  --  *      *       192.168.56.20      0.0.0.0/0          tcp dpt:80
Rule verified present.

(root@kali)~/SBT-DF203-Lab6
```

Part D - Capture Blocked Traffic and Rule Counters

sudo tshark -i eth0 -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -w evidence/http_blocked.pcapng &



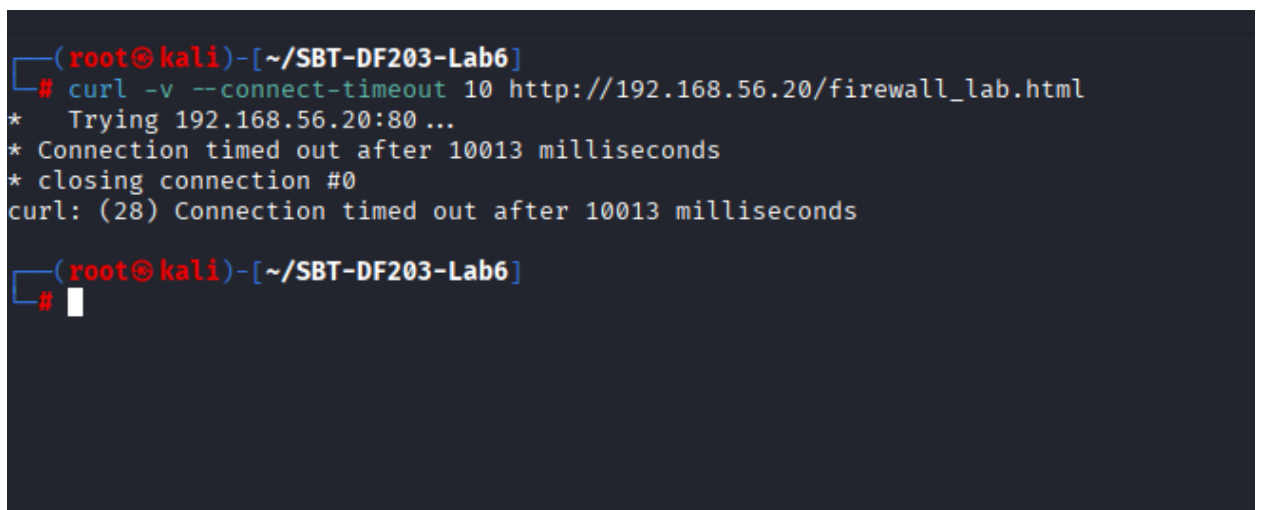
The screenshot shows a Kali Linux terminal window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The terminal is running as root at a kali machine in the directory ~/SBT-DF203-Lab6. The user enters the command `sudo tshark -i eth0 -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -w evidence/http_blocked.pcapng &`. The terminal output shows the command was executed successfully, with a PID of 26684. A warning message indicates the program was compiled against libxml 215 but is using older 214. The terminal then shows the command running in the background.

```
(root@kali)-[~/SBT-DF203-Lab6]
# sudo tshark -i eth0 -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -w
evidence/http_blocked.pcapng &
[1] 26684

(root@kali)-[~/SBT-DF203-Lab6]
# Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
8

[1] + done          sudo tshark -i eth0 -f 'host 192.168.56.20 and tcp port 80' -a d
uration:35 -w
(root@kali)-[~/SBT-DF203-Lab6]
#
```

curl -v --connect-timeout 10 http://192.168.56.20/firewall_lab.html



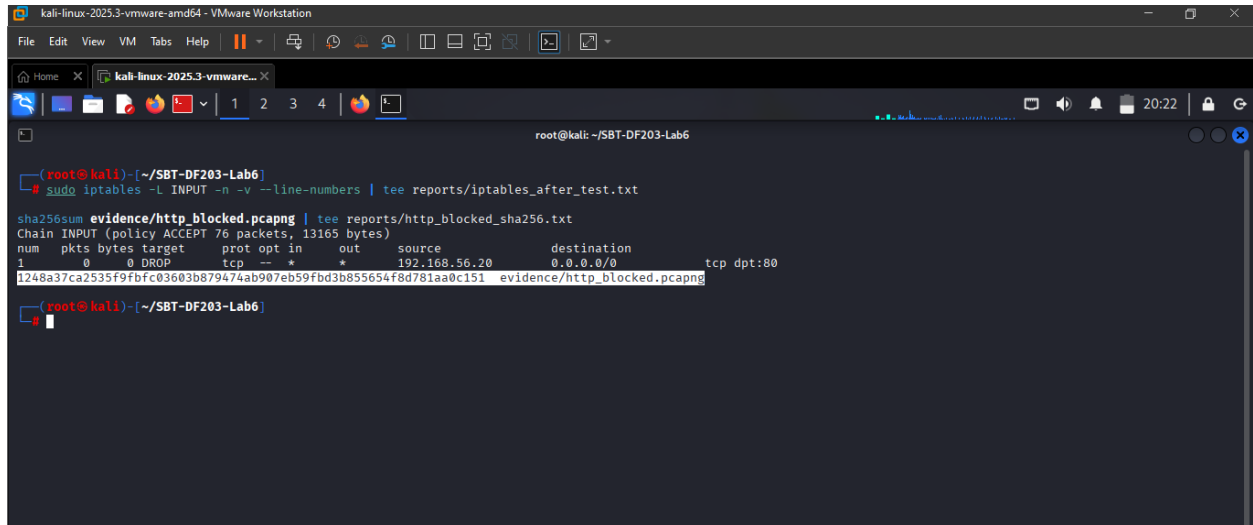
The screenshot shows a Kali Linux terminal window running the command `curl -v --connect-timeout 10 http://192.168.56.20/firewall_lab.html`. The output shows that the connection timed out after 10013 milliseconds.

```
(root@kali)-[~/SBT-DF203-Lab6]
# curl -v --connect-timeout 10 http://192.168.56.20/firewall_lab.html
* Trying 192.168.56.20:80 ...
* Connection timed out after 10013 milliseconds
* closing connection #0
curl: (28) Connection timed out after 10013 milliseconds

(root@kali)-[~/SBT-DF203-Lab6]
#
```

After capturing (Sha256sum)

1248a37ca2535f9fbfc03603b879474ab907eb59fbd3b855654f8d781aa0c151
(http_blocked.pcapng)



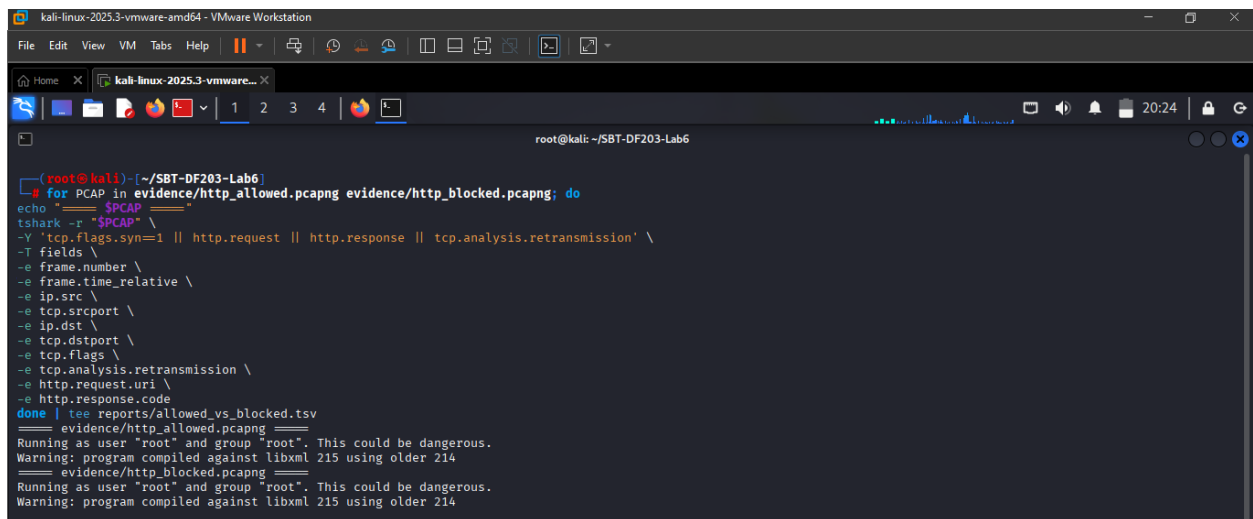
```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

(root@kali)~[~/SBT-DF203-Lab6]
# sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test.txt

Chain INPUT (policy ACCEPT 76 packets, 13165 bytes)
num  pkts bytes target     prot opt in     out     source                destination
1      0      0 DROP      tcp  --  *      *      192.168.56.20         0.0.0.0/0           tcp dpt:80
1248a37ca2535f9fbfc03603b879474ab907eb59fbd3b855654f8d781aa0c151  evidence/http_blocked.pcapng

(root@kali)~[~/SBT-DF203-Lab6]
#
```

Part E - Comparing Allowed and Blocked Captures



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

(root@kali)~[~/SBT-DF203-Lab6]
# for PCAP in evidence/http_allowed.pcapng evidence/http_blocked.pcapng; do
echo "==== $PCAP ====="
tshark -r "$PCAP" \
-Y 'tcp.flags.syn=1 || http.request || http.response || tcp.analysis.retransmission' \
-T fields \
-e frame.number \
-e frame.time_relative \
-e ip.src \
-e tcp.srcport \
-e ip.dst \
-e tcp.dstport \
-e tcp.flags \
-e tcp.analysis.retransmission \
-e http.request.uri \
-e http.response.code
done | tee reports/allowed_vs_blocked.tsv
==== evidence/http_allowed.pcapng ====
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
==== evidence/http_blocked.pcapng ====
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
```

Indicator	Allowed Capture	Blocked capture
Client SYN	Yes	Yes
Server SYN-ACK	Yes	No
TCP Handshake	Complete	Incomplete
HTTP GET	Visible	Not Visible

HTTP Response	Visible	Not Visible
Retransmissions	None	Present
Firewall Counter	No change	Increased
Curl Output	200 OK	Timeout

Part F - Remove the Rule and Restore Access

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

(root@kali)~[~/SBT-DF203-Lab6]
# sudo iptables -D INPUT -s "192.168.56.20" -p tcp --dport 80 -j DROP

(root@kali)~[~/SBT-DF203-Lab6]
# sudo iptables -C INPUT -s "192.168.56.20" -p tcp --dport 80 -j DROP || echo "Rule successfully removed"
iptables: Bad rule (does a matching rule exist in that chain?).
Rule successfully removed

(root@kali)~[~/SBT-DF203-Lab6]
# sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_restored.txt
Chain INPUT (policy ACCEPT 181 packets, 36788 bytes)
num  pkts bytes target     prot opt in     out     source            destination

(root@kali)~[~/SBT-DF203-Lab6]
#

```

Confirming access

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab6

(root@kali)~[~/SBT-DF203-Lab6]
# curl -v --connect-timeout 5 http://192.168.126.131/firewall_lab.html
* Trying 192.168.126.131:80 ...
* Established connection to 192.168.126.131 (192.168.126.131 port 80) from 192.168.126.131 port 55642
* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.126.131
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Sat, 19 Sep 2026 00:38:31 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 23:31:59 GMT
< ETag: "74-65bca4e8499ee"
< Accept-Ranges: bytes
< Content-Length: 116
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: IDRIS-SERVER</p></body></html>
* Connection #0 to host 192.168.126.131:80 left intact

(root@kali)~[~/SBT-DF203-Lab6]
#

```

HTTP/1.1 200 OK as Expected

Summary:

The lab successfully demonstrated the creation, verification and removal of a targeted iptables firewall rule. Network captures, firewall counters and application behaviour collectively confirmed that the rule effectively blocked HTTP traffic from the specified client while maintaining service availability for authorized hosts.

Forensic Interpretation Questions

- 1. Why does DROP commonly cause SYN retransmissions and a timeout?**
Because the firewall silently discards packets. The client receives no response and retransmits SYN packets until the connection attempt times out.
- 2. How would a REJECT rule differ in the packet capture and client output?**
- 3. Why are firewall counters valuable evidence?**
They prove that packets matched a specific firewall rule
- 4. Evidence that the web server was down instead of firewall blocked?**
No listener on ports 80, connection refusal and absence of HTTP responses.
- 5. What is the risk of deleting a rule by line number after other rules have changed?**
Rule positions can change after modifications.

